PHYSICS TESTING QUESTION BANK

Class XII — Question Compendium

SECTION A: MULTIPLE CHOICE QUESTIONS (MCQS)

Each question carries 1 mark. Select the most appropriate option from the choices given below.

Q1. In a region, the electric field is given by $\boldsymbol{E = 4x\hat{i}\ N/C}$. The potential difference between points A ($\boldsymbol{x = 1\ m}$) and B ($\boldsymbol{x = 3\ m}$), ($\boldsymbol{V_A - V_B}$), is:

(A) -16 V

(B) 16 V

(C) -8 V

(D) 8 V

Q2. In an unbiased p-n junction, at equilibrium, which of the following statements is true?

(A) Diffusion current is zero but drift current exists.

(B) Diffusion current exists but drift current is zero.

(C) Diffusion and drift currents are equal and opposite.

(D) Both the diffusion and drift currents exist but are unequal.

Q3. The reactance of a capacitor of capacitance $\boldsymbol{C}$ connected to an ac source of frequency $\boldsymbol{f}$ is '$\boldsymbol{X}$'. If the capacitance of the capacitor is doubled and the frequency of the source is tripled, the reactance will become:

(A) $\boldsymbol{X / 6}$

(B) $\boldsymbol{6X}$

(C) $\boldsymbol{2/3\ X}$

(D) $\boldsymbol{3/2\ X}$

Q4. A copper wire is stretched to increase its length by 1%. Then the change in its resistance is close to:

(A) 1%

(B) 4%

(C) -4%

(D) 2%

Q5. A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force F. If the distance of the charge is '$2r$', the force on the charge will be:

(A) $F / 16$

(B) $F / 8$

(C) $F / 4$

(D) $F / 2$

Q6. A conducting wire connects two charged metallic spheres A and B of radii r_1 and r_2 respectively. The distance between the spheres is very large compared to their radii. The ratio of electric fields, (E_A / E_B) at the surfaces of spheres A and B will be:

(A) r_1 / r_2

(B) r_2 / r_1

(C) r_1^2 / r_2^2

(D) r_2^2 / r_1^2

Q7. The focal length of a concave mirror in air is f. When the mirror is immersed in a liquid of refractive index $5/3$, its focal length will become:

(A) $5/3\ f$

(B) $3/5\ f$

(C) $2/3\ f$

(D) f

Q8. Consider a parallel circuit connection where the top branch contains a 12 V cell with 1 Ω internal resistance, and the bottom branch contains a 6 V cell with 0.5 Ω internal resistance connected with opposing polarities across parallel terminals A and B. The potential difference between points A and B is:

(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

Q9. A galvanometer of resistance G is converted into an ammeter of range 0 to I A. If the current through the galvanometer is 0.1% of I A, the total combined resistance of the ammeter circuit configuration is:

(A) $G / 999\ \Omega$

(B) $G / 1000\ \Omega$

(C) $G / 1001\ \Omega$

(D) $G / 100.1\ \Omega$

Q10. A charged particle gains a speed of 10^6 m/s when accelerated from rest through a potential difference of 10 kV. It enters a region of a uniform magnetic field of 0.4 T such that its velocity vector is perpendicular to the magnetic field vector. The radius of the circular path described by it is:

(A) 2.5 cm

(B) 5 cm

(C) 8 cm

(D) 10 cm

SECTION A: ASSERTION-REASONING QUESTIONS

For questions 11 to 16, two statements are given—one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer from the following options:
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(C) (A) is true, but (R) is false.
(D) Both (A) and (R) are false.

Q11.
Assertion (A): In a semiconductor diode, the thickness of the depletion layer is not fixed.
Reason (R): The thickness of the depletion layer in a semiconductor device depends upon many factors such as the biasing voltage state applied to the semiconductor.

Q12.
Assertion (A): The phase difference between any two points on a given wavefront is always zero.
Reason (R): All secondary source points distributed across a wavefront are located at an identical optical path distance from the source and thus oscillate in phase synchronization.

Q13.
Assertion (A): The induced electromotive force (emf) produced in a coil will be greater when the absolute magnitude of the magnetic flux linked with the coil is higher.
Reason (R): The induced emf produced inside a conducting loop is directly proportional to the total instantaneous magnetic flux linked with the circuit space.

Q14.

Assertion (A): Ideal plane and convex mirror systems cannot produce real focus images under any practical circumscribed experimental setup conditions.

Reason (R): A virtual image structure generated by a primary optics component can never serve as a valid virtual object to produce a real image output from a secondary boundary element.

Q15.

Assertion (A): When a structural bar made of pure copper is placed inside an external magnetic field region, the field flux lines get strongly concentrated within the internal volume of the bar.

Reason (R): Copper acts as a standard paramagnetic material block when subjected to external fields.

Q16.

Assertion (A): Binding energy is released when heavy atomic nuclei undergo fission or when light elemental nuclei undergo fusion processing loops.

Reason (R): For heavy nuclei, the binding energy per nucleon parameter increases monotonically with increasing atomic number Z, while for light elements it decreases with an increasing value of Z.

SECTION B: VERY SHORT ANSWER QUESTIONS (VSAQS)

Each question carries 2 marks. Answer the following questions precisely with necessary formulas or clear point-wise steps.

Q17. The threshold voltage of a silicon diode is 0.7 V. It is operated by connecting the diode in series with a battery of V volts and a resistor of 1000 Ω. Find the required value of V when the active current drawn through the circuit is 15 mA.

Q18. Two coherent monochromatic light beams possessing intensities I and 4I undergo superposition with each other. Determine the quantitative ratio of the maximum and minimum intensities observed within the resulting interference pattern profile.

Q19. What is the characteristic order of magnitude of the drift velocity of free electrons inside a typical metal conductor under standard fields? Deduce the foundational mathematical relation between the current flowing through a conductor and the drift velocity of electrons within it.

Q20. Plot a neat current-voltage (I-V) characteristic curve for the semiconductor material gallium arsenide (GaAs). On the generated graph layout, explicitly mark: (a) the region where the dynamic operational resistance becomes negative, and (b) the region where Ohm's law is obeyed.

Q21. A plane circular conducting coil is rotated about its vertical diameter layout with a constant angular speed $\boldsymbol{\omega}$ inside a uniform horizontal magnetic field. Initially, the plane of the coil is parallel to the field vector. Draw dual synchronous plots showing the variation of: (a) the magnetic flux linked with the coil, and (b) the emf induced in the coil, as functions of the phase parameter ($\boldsymbol{\omega t}$).

Q22. The ground state energy of a hydrogen atom is given as -13.6 eV. Calculate the precise potential energy value and kinetic energy value of the bound electron when the atom is excited to its third excited state configuration.

Q23. A conducting wire of fixed total length L is bent sequentially into: (i) a square coil containing N total turns, and (ii) a circular coil containing N total turns. The coil structure in both experimental cases is free to pivot about a vertical axis coinciding with its plane inside a uniform, horizontal external magnetic field. Calculate the ratio of the maximum torque acting on the square coil configuration to that acting on the circular coil layout when carrying identical steady currents.

Q24. Consider a neutron of mass m possessing an absolute kinetic energy E, and a photon sharing the exact same energy value E. Let $\boldsymbol{\lambda_n}$ represent the de Broglie wavelength of the neutron and $\boldsymbol{\lambda_p}$ be the structural wavelength of the photon. Derive a simplified analytical expression for the ratio $\boldsymbol{\lambda_n / \lambda_p}$ in terms of the system parameters.

Q25. A mixed light beam consisting of two wavelengths, 400 nm and 600 nm, is utilized to normally illuminate a single slit having an aperture width of 1 mm. Calculate the minimum linear distance from the central maximum on a screen placed 1.5 m away where the dark fringes produced by both individual wavelengths completely overlap.

Q26. A steady current of 5 A passes along the +X coordinate direction through a long straight wire lying coincident with the X-axis. Find the vector magnetic field $\boldsymbol{B}$ generated at a spatial coordinate point located at position vector $\boldsymbol{r = (3\hat{i} + 4\hat{i})\ m}$ due to a 1 cm segment element of the wire centered symmetrically at the origin.

SECTION C: SHORT ANSWER QUESTIONS (SAQS)

Each question carries 3 marks. Provide clear analytical explanations, derivations, or step-wise calculations.

Q27. Identify the specific radiation segments of the electromagnetic spectrum used in: (i) radar navigation systems for aircraft coordination, (ii) clinical diagnostic imaging of internal skeletal anatomy anomalies, and (iii) geostationary satellite communication links. Provide one additional distinct secondary application for each identified radiation type.

Q28. State the foundational postulates of Bohr's semi-classical model of the hydrogen atom. Utilizing these postulates, derive the mathematical expression for the radius of the stable $\boldsymbol{n^{th}}$ orbit of an electron revolving around the central nucleus.

Q29. Distinguish clearly between an n-type semiconductor and a p-type semiconductor material block on the basis of their respective valence and conduction energy band structures at a thermal equilibrium state temperature of $\boldsymbol{T > 0\ K}$. Explicitly mark the positions of the donor and acceptor impurity levels on the drawn layout.

Q30. An alternating voltage source represented by $V = V_0 \sin \omega t$ is applied across an ideal inductor possessing self-inductance L. Derive the mathematical expression for the instantaneous current flowing through the circuit. Prove analytically that the current wave lags behind the voltage wave by a phase angle of $\pi / 2$ radians, and construct the corresponding vector phasor diagram layout.

Q31. Define the terms 'work function' and 'stopping potential' within the context of photoelectric emission phenomena. Plot a combined graph showing the variation of stopping potential with the frequency of incident radiation for two independent photosensitive metallic surfaces A and B, given that the work function of metal A is greater than that of metal B. State how the value of Planck's constant can be derived from this graph profile.

Q32. State the two electrical network circuit analysis rules formulated by Kirchhoff. Apply these rules directly to calculate the steady-state current flowing through each independent internal branch of a parallel network where a 12 V cell with 1 Ω internal resistance is wired in parallel with a 6 V cell having 2 Ω internal resistance across an external load of 5 Ω.

Q33. A Young's double-slit experiment setup is illuminated with a composite light source consisting of a known reference wavelength of 520 nm and an unknown operational wavelength λ. The system generates overlapping interference fringe fields on a distant screen such that the fourth bright fringe corresponding to the unknown wavelength falls exactly coincident on the position of the fifth bright fringe of the known reference source. Calculate the value of the unknown wavelength λ.

Q34. The refractive index of the material of a triangular glass prism is given as $\sqrt{2}$. If the refracting vertex angle of the prism is $60°$, determine: (i) the angle of minimum deviation experienced by a monochromatic beam passing symmetrically through the prism, and (ii) the required angle of incidence at the first boundary face to achieve this minimum deviation state.

Q35. A parallel plate air capacitor has an initial capacitance value of 9 pF across a uniform plate separation distance d. The inter-plate space volume is now modified by introducing two distinct dielectric slabs: one slab has a dielectric constant $K_1 = 3$ and a linear thickness of $d / 3$, while the remaining space thickness of $2d / 3$ is completely filled by a second slab possessing a dielectric constant $K_2 = 6$. Calculate the new modified effective capacitance of the system.

Q36. A long, straight, solid cylindrical copper conductor possessing an overall radius R carries a steady current I that is distributed uniformly across its cross-sectional area. Apply Ampere's Circuital Law to derive explicit expressions for the magnetic field magnitude B at a radial distance r from the cylinder's central axis for: (a) inside points ($r < R$), and (b) outside points ($r > R$). Plot a continuous graph showing the variation of B from $r = 0$ out to points where $r >> R$.

SECTION D: LONG ANSWER QUESTIONS (LAQS)

Each question carries 5 marks. Comprehensive solutions detailing equations, schematic diagrams, and thorough derivations are required.

Q37. (a) Generalize the mathematical relation for refraction across a single curved spherical interface separating two media of distinct refractive indices. Utilizing this, derive the step-by-step expression for the Lens Maker's Formula for a thin biconvex lens element.
(b) Define the optical power parameters of a lens element and explicitly state its standard SI unit.

Q38. (a) Analyze the microscopic physical source of the mechanical force acting on a moving charge carrier segment passing through a conductor loop placed inside an external magnetic field field zone.
(b) Deduce the complete analytical expression for the mechanical force acting per unit length between two infinitely long, straight, parallel current-carrying conductors separated by a distance d in vacuum. Use the derived expression to formally define the standard base SI unit 'ampere'.

Q39. (a) State Huygens' core geometric principle of wavefront propagation and secondary wavelets tracking over time.
(b) A plane wavefront is incident at a non-zero angle i on a perfectly smooth reflecting plane interface. Construct a clean geometrical layout showing the generation of the corresponding reflected wavefront using Huygens' construction method. Utilizing the drawn layout, prove the fundamental law of reflection.

Q40. (a) Define the electromagnetic parameter 'mutual inductance' tracking between a closed pair of independent inductor coils, and state its standard SI units.
(b) Derive the comprehensive mathematical expression for the mutual inductance coefficient linking two long, coaxial solenoids of identical total length l, given that the inner solenoid has radius r_1 and N_1 total turns, while the outer surrounding solenoid features radius r_2 and N_2 total turns.

Q41. With the help of a detailed, clearly labelled ray diagram, describe the construction, alignment adjustment, and geometric working of a standard Compound Microscope system adjusted such that the final virtual image is generated exactly at the least distance of distinct vision (D). Derive the complete mathematical expression for the total linear magnifying power of the system under this specific operational condition.

Q42. A thin, isolated spherical conducting shell S_1 encloses a bound volume containing three individual localized point charges: $q_1 = 3\ \mu C$, $q_2 = -2\ \mu C$, and $q_3 = 9\ \mu C$. This entire configuration space is subsequently enclosed by a larger concentric thin conducting shell element S_2. A standalone point charge Q is placed in the hollow space interface separating the two shells. If experimental testing reveals that the net electrostatic flux passing through the outer bounding surface S_2 is exactly four times the net flux tracked passing through the inner shell surface S_1, calculate the value of the unknown intermediate charge Q.

Q43. Three thin convex lenses L_1, L_2, and L_3, each possessing an identical individual focal length value of +40 cm, are mounted coaxially along a straight bench line. The separation distance between the first lens L_1 and the middle lens L_2 is fixed at 120 cm, while the separation distance between the middle lens L_2 and the final lens L_3 is set to 20 cm. A point object is positioned on the axis at a distance of 80 cm to the left of the first lens element L_1. Determine the final position coordinates of the image formed by the multi-component combination relative to the position of the original target object.

Q44. (a) State the working electrical principle of a standard potentiometer measurement circuit layout.
(b) A parallel electrical combination composed of two independent cells of electromotive forces E and $3E$, with matching internal resistances R each, is wired across an external load resistance branch of magnitude $2R$. Derive the expression for, and calculate the steady electric current that passes through the external load resistor branch.

Q45. (a) A parallel monochromatic light beam falls normally onto a single slit having an aperture width parameter 'a' to produce a standard Fraunhofer diffraction pattern on a screen positioned at a distance D. Explain clearly: (i) the physical mechanics governing the formation of secondary maxima and minima zones, and (ii) why the secondary maxima peaks go on becoming weaker and weaker in intensity as their order integer n increases.
(b) Detail two distinct physical and structural points of difference to contrast an interference field due to a double-slit layout from a single-slit diffraction field configuration.

Q46. (a) Define the self-inductance parameter of an electrical component. Derive the mathematical expression for the total internal magnetic energy stored within an inductor of self-inductance L when the current has been built up to a steady state value I.
(b) The currents passing through two independent isolated inductor components of self-inductances 10 mH and 20 mH are forced to increase at the exact same uniform rate with respect to time. Sketch distinct comparative graphs showing the variation of: (i) the induced emf magnitude as a function of the current ramp rate, and (ii) the total stored magnetic energy as a function of the instantaneous current value.
(c) The current passing through an inductor is increased uniformly from 0 to 2 A over a time window of 40 s. If an internal emf of 5 mV is induced during this transition period, calculate the total magnetic flux linked with the inductor at the specific intermediate instant $t = 10 ext{ s}$.

SECTION E: CASE STUDY-BASED QUESTIONS

Each question block carries 4 marks. Read the context passages carefully and answer all the sub-questions that follow.

Q47. Passage: A parallel plate capacitor has two parallel plates which are separated by an insulating medium like air, mica, etc. When the plates are connected to the terminals of a battery, they get equal and opposite charges and an electric field is set up in between them. This electric field between the two plates depends upon the potential difference applied, the separation of the plates and nature of the medium between the plates.

Sub-questions:

1. The electric field between the plates of a parallel plate capacitor is E. If the separation between the plates is doubled and simultaneously the applied potential difference between the plates is reduced to half of its initial value, calculate the new value of the electric field between the plates.

2. A constant electric field is to be maintained between the two plates of a capacitor whose separation d changes with time. Which specific mathematical function profile relates the potential difference V that must be applied to the plates to the instantaneous separation distance d?

3. A material of dielectric constant K is completely filled inside a parallel plate capacitor of initial capacitance C. Determine the new value of its capacitance.

Q48. Passage: Strontium titanate is a rare oxide, a natural mineral found in Siberia. It is used as a substitute for diamond because its refractive index and critical angle are 2.41 and 24.5° respectively, which are approximately equal to the refractive index and critical angle of diamond. It has all the properties of diamond. Even an expert jeweller is unable to differentiate between diamond and strontium titanate. A ray of light is incident normally on one face of an equilateral triangular prism ABC made of strontium titanate.

Sub-questions:

1. Trace the path of the ray showing its passage through the equilateral triangular prism and state whether the ray will successfully emerge from the second face or undergo Total Internal Reflection (TIR).

2. Calculate the exact velocity of light as it propagates inside the strontium titanate prism medium.

3. Define total internal reflection of light and state the two essential physical boundary conditions required for its occurrence.

Q49. Passage: Einstein explained the photoelectric effect on the basis of Planck's quantum theory, where light travels in the form of small bundles of energy called photons. The energy of each photon is given by $\boldsymbol{hv}$, where $\boldsymbol{v}$ is the frequency of incident light and $\boldsymbol{h}$ is Planck's constant. The number of photons in a beam of light determines the intensity of the incident light. A photon incident on a metal surface transfers its total energy to a free electron in the metal. A part of this energy is used in ejecting the electron from the metal and is called its work function. The rest of the energy is carried by the ejected electron as its kinetic energy.

Sub-questions:

1. Detail which specific experimental coordinate plot can be utilized to graphically evaluate the exact numerical value of Planck's constant.

2. Monochromatic red light, yellow light, and blue light beams sharing identical total intensity values are incident on a target metal surface sequentially. If $\boldsymbol{K_R}$, $\boldsymbol{K_Y}$, and $\boldsymbol{K_B}$ represent the maximum kinetic energy of the emitted photoelectrons respectively, arrange them in increasing order.

3. When the frequency of the incident light is increased above the threshold value without altering the total beam intensity, analyze the effect on the maximum saturation photocurrent.

Q50. Passage: The electric potential (V) and electric field (E) are closely related concepts in electrostatics. They are related by the gradient equations, meaning that the electric field points in the direction of decreasing potential and its magnitude is the rate of change of potential with distance. Consider a pair of square conducting plates having sides of length 0.05 m arranged parallel to each other in the x-y plane. They are separated by 0.01 m along the z-axis and connected across a 200 V DC power supply. An electron enters with a horizontal velocity of $\boldsymbol{3 \times 10^7\ m/s}$ symmetrically through the space midway between the two plates. Neglect any gravitational field effects.

Sub-questions:

1. Calculate the magnitude and coordinate direction vector of the uniform electric field generated in the region between the plates.

2. Determine the vector acceleration experienced by the electron while it moves within the inter-plate space.

3. Calculate the total time interval during which the electron remains within the plates, and find its corresponding vertical deflection displacement coordinate at the exit edge.

Q51. Passage: A galvanometer is an instrument used to show the direction and strength of the current passing through it. It essentially works on the fact that a current-carrying coil experiences a deflecting torque when placed in a magnetic field. A spring attached to the coil provides a counter torque. In equilibrium, the deflecting torque is balanced by the restoring torque of the spring: $NBAI = k\varphi$. A small resistance, called a shunt, of a suitable value is connected in parallel with the galvanometer to convert it into an ammeter of a desired range. By connecting a high resistance in series, a galvanometer can be converted into a voltmeter.

Sub-questions:

1. A galvanometer is converted into a voltmeter of range (0-V) by connecting a resistance R_1 in series. If R_1 is replaced by a series resistance R_2, the voltmeter range scales to (0-2V). Derive an expression for the internal resistance of the galvanometer coil (G) in terms of R_1 and R_2.

2. A current of 5 mA flows through a galvanometer whose suspended coil contains 100 turns, each turn possessing an area of cross-section of $18\ cm^2$. If it is suspended in a uniform radial magnetic field of 0.20 T, calculate the deflecting torque acting on the coil.

3. A galvanometer of internal resistance $6\ \Omega$ shows full-scale deflection for a current of 0.2 A. Calculate the resistance of the shunt required to convert it into an ammeter capable of measuring a full range from 0 to 5 A.

Q52. Passage: Complex electrical networks containing multiple junctions, cells, and resistors cannot be easily simplified using Ohm's law alone. We can find the currents through and potential differences across different branches of such circuits using Kirchhoff's rules. Kirchhoff's first rule (Junction Rule) is a statement of conservation of electric charge, while his second rule (Loop Rule) is based on the law of conservation of energy. Consider a network mesh where a primary branch loop contains an EMF source $E = 6.0\ V$ with an internal resistance $r = 1.0\ \Omega$ connected in a circuit that branches into three parallel resistors: $R_1 = 10\ \Omega$, $R_2 = 5.0\ \Omega$, and $R_3 = 5.0\ \Omega$.

Sub-questions:

1. Identify which node points within a standard parallel multi-mesh network share matching potential values under steady-state conditions.

2. Set up the loop equations and calculate the current passing through the branch loop containing resistance R_1.

3. Calculate the potential difference across resistance R_3 and determine the total power dissipated across resistance R_2 under steady state flow.